# The Echo Chamber Effect: A Comparative Analysis of Response Homogeneity in Modern LLMs

Tanishq Mann
B.Tech CSE (AI/ML)
Sushant University
tanishq.220btccse024@sushantuniversity.edu.in

Hardhik Sharma
B.Tech CSE (AI/ML)
Sushant University
hardhik.220btccse009@sushantuniversity.edu.in

Raghavender Sharma
B.Tech CSE (AI/ML)
Sushant University
raghavender.220btccse003@sushantuniversity.edu.in

Dr. Nitin Malik
Dean SET
Sushant University
deanset@sushantuniversity.edu.in

***Abstract*—Large Language Models (LLMs) are increasingly evaluated on factual correctness, instruction-following fidelity, and alignment safety, yet these metrics under-characterize a critical emergent failure mode in open-ended generation: response manifold contraction under repeated sampling. This work formalizes and empirically probes that contraction as the *Echo Chamber Effect*, i.e., the tendency of modern LLMs to produce semantically convergent outputs despite stochastic decoding, prompt openness, and lexical variability. Motivated by recent literature on open-ended model collapse, we now structure the study as a two-phase comparative framework. Phase 1 investigates *intra-model temperature ablation*, measuring whether semantic spread meaningfully changes as decoding temperature increases, including $T = 1.0$. Phase 2 extends the same embedding-space methodology to *inter-model API evaluation* across GPT-4.1, Llama-3.3, and Gemini-2.5.**

**Across both phases, the pipeline operationalizes generation diversity as a geometric property of the induced semantic distribution: for each prompt family, we sample multiple generations under controlled decoding settings, encode responses using HuggingFace `all-MiniLM-L6-v2` sentence embeddings, and quantify pairwise semantic affinity via cosine similarity over the resulting dense vectors. To inspect neighborhood topology and latent mode occupancy, we project high-dimensional embeddings into two-dimensional manifolds using t-SNE with fixed random seed and bounded perplexity, then stratify points by prompt identity, temperature regime, and model family. Phase 1 shows that increased token-level entropy does not linearly translate into semantic dispersion: similarity matrices exhibit block-structured off-diagonal concentration, and t-SNE projections show compact, recurring clusters with only sparse outlier trajectories. This indicates that decoding randomness primarily perturbs surface realization while preserving shared latent attractors, producing high intra-model semantic autocorrelation even in nominally creativity-oriented prompts.**

**Phase 2 demonstrates that the same contraction pattern extends across model families rather than remaining confined to a single architecture. In particular, our inter-model API results show a 78.63% semantic agreement between the OpenAI and Meta model families, specifically between GPT-4.1 and Llama-3.3, while Gemini-2.5 serves as a third comparative baseline in the broader evaluation. The technical contribution of the study is therefore twofold: (i) a modular, MLOps-ready evaluation scaffold that unifies generation, embedding, similarity analysis, and manifold diagnostics across intra- and inter-model settings; and (ii) quantitative evidence that semantic mode concentration persists both within a model under temperature variation and across modern frontier APIs. At a broader systems level, the study highlights why future alignment and decoding research must jointly optimize for correctness, safety, and semantic pluralism to avoid long-term homogenization of machine-mediated ideation.**

## I. Introduction

The contemporary trajectory of Artificial Intelligence (AI), especially within Natural Language Processing (NLP), has moved from narrow task-optimized systems toward broadly capable generative architectures that increasingly mediate human communication, learning, programming, design, and ideation. In this transition, evaluation priorities have remained disproportionately centered on correctness, safety compliance, and instruction adherence, whereas diversity of semantic exploration has been treated as a secondary or emergent property. This asymmetry has produced a measurement blind spot: models that appear robust under conventional benchmarks may still exhibit severe compression in conceptual output space when repeatedly queried in open-ended settings. The present work positions this compression as a systems-level risk rather than a stylistic artifact. Specifically, when large language models (LLMs) repeatedly return semantically proximate responses across prompts intended to elicit plurality, users are exposed to a narrowed idea manifold that may recursively shape thought patterns, engineering decisions, and research pathways.

In practical computing ecosystems, this risk is non-trivial. Students use LLMs to generate solution approaches, engineers use them for architectural alternatives, developers use them for code pattern synthesis, and researchers use them for hypothesis framing. If model outputs converge toward the same latent templates despite lexical variation, then the epistemic diversity of computational practice is attenuated. Over time, such attenuation can create feedback loops where homogeneous model outputs influence homogeneous human artifacts, which then re-enter future training corpora and reinforce the same representational attractors. Therefore, the challenge investigated in this project is not merely whether LLMs can produce valid

language, but whether they preserve enough semantic spread to support genuine cognitive and computational pluralism.

This Introduction develops that argument through four focused layers: the architectural origins of modern generative systems, the alignment-era constraints introduced by Reinforcement Learning from Human Feedback (RLHF), the formal definition of the *Artificial Hivemind* phenomenon, and the precise research objectives operationalized through our Python embedding-analysis pipeline.

### A. 1.1 The Evolution of Generative AI and Natural Language Processing

The evolution of NLP can be understood as a sequence of representational paradigm shifts. Early statistical methods relied on sparse symbolic features and local co-occurrence statistics. N-gram models, Hidden Markov Models, and feature-engineered classifiers established foundational progress but were limited by short-context assumptions and brittle generalization under distributional shift. The deep learning wave introduced dense embeddings and sequence models, especially recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) architectures, enabling richer contextualization; however, recurrence imposed computational bottlenecks and degraded long-range dependency modeling due to vanishing signal paths and limited parallelism.

The transformer architecture represented a discontinuity rather than an incremental improvement. Self-attention replaced recurrence with direct token-to-token relational modeling, allowing each position to condition on all others in a single layer. Multi-head attention further decomposed representational focus into parallel relational subspaces, while positional encoding preserved order information absent in permutation-invariant attention primitives. In scaled settings, this design unlocked both hardware-efficient training and a qualitatively different capability profile: broad transfer learning, in-context adaptation, and emergent behavior across heterogeneous tasks.

From an engineering perspective, transformers transformed NLP from task-specific modeling into foundation-model pretraining followed by instruction adaptation. Massive unsupervised next-token prediction on internet-scale corpora produced versatile latent language representations, while downstream fine-tuning and prompt-based conditioning exposed those representations for practical use. The resulting systems moved beyond classical NLP boundaries and became general-purpose cognitive interfaces for software development, documentation synthesis, requirements engineering, mathematical explanation, and code generation.

However, scale and fluency also obscured a critical distinction: lexical richness is not equivalent to semantic diversity. A model can generate many different surface realizations that map to a small set of latent conceptual templates. In other words, apparent variation at the token level may coexist with manifold contraction at the embedding level. This distinction becomes especially salient in open-ended tasks such as brainstorming, metaphor generation, design alternatives, and exploratory question answering, where output plurality is not peripheral but central to utility.

The computational mechanics of transformers can partially explain this phenomenon. During pretraining, the objective encourages high-likelihood continuations conditioned on broad corpus priors. This likelihood geometry can produce steep basins around culturally dominant or statistically frequent narrative frames. At inference time, decoding strategies (temperature scaling, top-$k$, top-$p$ sampling) perturb traversal within this learned distribution but do not necessarily escape basin-level attractors. Consequently, as model size and alignment sophistication increase, one may observe a paradox: better coherence and safety yet reduced conceptual divergence under repeated generation.

For computer science education and engineering practice, this historical arc has a direct implication. The very architectures that democratized high-quality text generation also introduced infrastructure-level dependence on a few dominant semantic priors. Without explicit diversity diagnostics, these priors can silently propagate through programming pedagogy, software specification language, and design ideation patterns, producing convergence where exploration is expected.

### B. 1.2 The Paradigm Shift to RLHF and Alignment Protocols

As LLMs moved from research artifacts to deployed systems, the optimization target expanded from predictive performance to behavioral acceptability. Reinforcement Learning from Human Feedback (RLHF), constitutional alignment methods, preference-model fine-tuning, and rule-constrained decoding became central to production pipelines. These methods substantially improved helpfulness, harmlessness, and response reliability, reducing toxic failure cases and instruction refusal inconsistencies. Yet, from a variance perspective, alignment introduces a structural regularization pressure that can compress output diversity.

RLHF pipelines typically involve three stages: supervised instruction tuning, reward model training from human preference pairs, and policy optimization (often via proximal methods) against learned reward signals. Each stage tends to privilege responses that satisfy generalized human-annotator preference distributions. While beneficial for safety and quality, this process can reduce tail exploration. Outputs that are unconventional, stylistically idiosyncratic, or semantically risky may be down-weighted even when they are valid and potentially creative. Over many optimization iterations, policy behavior is pulled toward high-reward consensus regions.

In high-stakes deployment this is rational. Organizations prioritize predictability, low legal risk, and minimal harmful content. However, for open-ended computational creativity, consensus-optimized behavior can produce an unintended collapse toward normative templates. The model learns not only *what is safe*, but also *what is most generally acceptable*, and these are not equivalent objectives in ideation-intensive contexts. A response space shaped primarily by average preference can under-represent minority epistemic styles, unconventional solution pathways, and domain-specific creative heuristics.

Alignment protocols also interact with decoding in nonlinear ways. Temperature amplification increases token-level entropy but does not erase reward-shaped policy geometry. If the reward landscape has narrow high-value corridors, stochastic decoding explores nearby variants of those corridors rather than globally diverse conceptual regions. This explains why higher temperature can yield greater phrasing variability without substantial semantic displacement. In practical terms, the user perceives different wording while receiving similar ideas.

Another technical factor is instruction hierarchy stabilization. Modern aligned LLMs encode strong priors for answer structure: preamble, clarification, balanced framing, risk disclaimer, and actionable conclusion. Such scaffolding is useful for reliability but can further homogenize discourse topology. In coding and software design tasks, this may manifest as recurring architecture recommendations, repeated framework choices, and convergence to standard implementation motifs regardless of prompt diversity. Over time, collective dependence on such outputs may reduce experimentation in student projects, hackathon prototypes, and even early-stage research conceptualization.

Thus, the alignment era should not be interpreted as a binary trade-off between safety and capability; rather, it introduces a multi-objective optimization problem where correctness, safety, usefulness, and semantic pluralism must be jointly balanced. The central concern of this paper emerges in this context: when alignment suppresses variance beyond beneficial regularization, an invisible form of algorithmic monoculture can arise.

### C. 1.3 Defining the 'Artificial Hivemind' Phenomenon

We define the *Artificial Hivemind* as a measurable condition in which repeated generations from one or more LLMs concentrate in a narrow semantic neighborhood despite prompt openness and stochastic decoding. This condition is distinct from ordinary consistency. Healthy consistency preserves factual reliability while allowing conceptual branching; hivemind behavior preserves stylistic fluency but collapses branching structure.

Formally, let $p \in \mathcal{P}$ denote an open-ended prompt and let $\{r_{p,i}\}_{i=1}^{n}$ be independently sampled responses. Let $\phi(\cdot)$ map text to embedding vectors in $\mathbb{R}^d$. The response set for a prompt induces a similarity matrix

$$S_{ij}^{(p)} = \frac{\phi(r_{p,i})^\top \phi(r_{p,j})}{\|\phi(r_{p,i})\|_2 \, \|\phi(r_{p,j})\|_2}. \tag{1}$$

If the model exhibits robust semantic diversity, off-diagonal elements of $S^{(p)}$ should be broadly dispersed. Under hivemind dynamics, off-diagonal mass shifts toward high similarity regimes, often accompanied by visible cluster compaction in low-dimensional projections.

Importantly, this is not merely an intra-model issue. Inter-model homogeneity occurs when distinct architectures, sizes, or providers generate responses that occupy overlapping semantic basins for the same open-ended query. In such cases, ecosystem-level diversity degrades even when users switch models, creating an illusion of choice without substantive conceptual variation.

From a computer science standpoint, the threat surface is broad:

- **Algorithmic design convergence:** repeated recommendation of similar data structures, optimization strategies, or system blueprints can reduce exploratory engineering.
- **Educational homogenization:** students using model assistance may submit structurally similar reasoning and implementation patterns, weakening independent problem-solving development.
- **Research ideation narrowing:** literature synthesis and hypothesis generation may over-prioritize dominant narratives, suppressing unconventional yet valuable directions.
- **Software monoculture risk:** recurrent coding motifs can amplify shared vulnerabilities, anti-pattern propagation, and architectural fragility.

Creative homogenization is therefore not only an aesthetic concern; it is a systems concern tied to innovation capacity, resilience, and long-term epistemic diversity in computational fields. If LLM-mediated workflows become default infrastructure, even modest reductions in semantic spread can accumulate into significant collective effects. The "echo chamber" metaphor is apt because each model output conditions subsequent human prompts, and subsequent prompts often condition future model outputs via data feedback pipelines. This recursive coupling creates a closed-loop dynamical system that can amplify convergence unless diversity-aware interventions are explicitly introduced.

### D. 1.4 Problem Statement and Research Objectives

Given the above context, the primary problem addressed in this project is:

> *Do modern generative language models, when sampled repeatedly in open-ended settings, exhibit statistically and geometrically significant semantic homogenization both within a single model under temperature variation and across different frontier model families, and can this homogenization be demonstrated through a reproducible embedding-space pipeline?*

To answer this question, our Python pipeline is designed not as a one-off visualization script but as a modular two-phase evaluation framework with explicit analytical objectives. Let $\mathcal{R}$ denote the full response corpus generated across prompts, temperature conditions, and model families. Each response is embedded via `all-MiniLM-L6-v2` to produce vectors $\mathbf{e}_1, \ldots, \mathbf{e}_N$. We then evaluate homogeneity through complementary lenses:

1) **Phase 1 — Intra-model temperature ablation:** construct cosine similarity matrix $\mathbf{S}$ and inspect whether off-diagonal concentration persists as decoding temperature varies, especially from low temperature to $T = 1.0$.

2) **Phase 1 — Local topology analysis:** apply t-SNE to map high-dimensional neighborhoods and detect whether repeated generations from the same model remain confined to compact semantic clusters despite temperature changes.
3) **Phase 2 — Inter-model API evaluation:** compare response manifolds across GPT-4.1, Llama-3.3, and Gemini-2.5 to test whether different frontier APIs converge toward shared semantic basins.
4) **Cross-phase prompt-conditioned behavior:** examine whether different prompt intents still collapse into limited semantic basins within and across model families.

The pipeline is specifically intended to test the following hypotheses:

- **H1 (Intra-model concentration):** For open-ended prompts, a significant fraction of response pairs will exhibit high cosine similarity, indicating limited semantic spread.
- **H2 (Temperature-insensitive contraction):** Increasing decoding temperature to $T = 1.0$ will not proportionally increase embedding-space dispersion.
- **H3 (Cluster persistence):** Dimensionality-reduced projections will reveal recurrent, compact semantic clusters rather than uniformly distributed manifolds.
- **H4 (Inter-model convergence):** GPT-4.1, Llama-3.3, and Gemini-2.5 will exhibit measurable semantic agreement under identical open-ended prompts, indicating that homogeneity is not purely model-specific.

Operationally, the objective is to transition from anecdotal claims ("responses feel similar") to measurable evidence grounded in vector geometry. This is crucial for engineering rigor: actionable mitigation strategies require quantifiable diagnostics. Under this two-phase design, Phase 1 establishes whether semantic contraction survives temperature ablation within a single model, while Phase 2 tests whether the same effect persists across major API-accessible model families. Our current Phase 2 results already indicate a 78.63% semantic agreement between the OpenAI and Meta systems, specifically GPT-4.1 and Llama-3.3, providing explicit numerical evidence that echo-chamber behavior extends beyond isolated intra-model sampling effects. If homogeneity is confirmed at scale, subsequent work can evaluate interventions such as diversity-aware decoding, prompt perturbation ensembles, multi-model response aggregation with de-correlation objectives, and reward-shaping approaches that preserve safety without collapsing semantic plurality.

Finally, this introduction frames a broader research imperative for AI/ML within computer science: future evaluation frameworks must treat *semantic diversity* as a first-class systems metric alongside accuracy and alignment. In environments where LLMs increasingly co-author code, documents, and ideas, preserving variation is foundational to innovation. Without such safeguards, the field risks drifting toward high-fluency uniformity—an efficient but epistemically brittle mode of machine-mediated intelligence.

## II. Literature Review

The literature surrounding collapse phenomena in generative systems has evolved into two related but analytically distinct traditions. The first concerns *model collapse* in recursive data-generation loops, where training distributions degrade when synthetic outputs are repeatedly re-ingested as supervisory signal. The second concerns *mode collapse* (or semantic homogenization) at inference time, where a model or model family generates low-diversity responses across open-ended prompts despite stochastic decoding. Although both phenomena involve contraction of effective support over an underlying distribution, they occur at different points in the lifecycle of AI systems: one in training-data dynamics and one in deployment-time generation behavior. This section synthesizes both literatures and positions the uploaded foundational paper, *Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)*, as the primary empirical anchor for our project.

### *A. 2.1 Traditional Model Collapse: Recursive Synthetic Data Degradation*

Traditional model-collapse literature, especially work on recursive training, argues that generative models trained on model-generated data progressively lose support over low-probability but semantically important regions. In practical terms, rare events, minority expressions, and distributional tails are attenuated after repeated synthetic retraining cycles. The commonly cited intuition is distributional contraction: if each generation round approximates high-density regions of a prior model, then repeated retraining can amplify central modes and erase fine-grained variability in tails. The consequence is not merely reduced novelty, but representational amnesia.

Formally, if $P_0(x)$ is the original human data distribution and $G_t$ generates synthetic samples used to form $P_{t+1}(x)$, then under imperfect generation and filtering,

$$P_{t+1}(x) \propto \alpha P_t(x) + (1-\alpha)\tilde{G}_t(x), \tag{2}$$

where $\tilde{G}_t$ under-represents rare-support regions. Over iterations, KL divergence between $P_t$ and the true human manifold can increase in tail-sensitive directions even when average likelihood metrics appear stable.

This training-time collapse is deeply relevant to long-term AI sustainability: if large-scale corpora become increasingly synthetic, future models may inherit compressed cultural and linguistic manifolds. However, this paradigm does not directly measure a user-facing question central to deployed LLMs: *given a fixed trained model, does repeated open-ended querying already produce low-diversity semantics in real usage?* That question motivates the second literature stream.

### *B. 2.2 Mode Collapse in Open-Ended LLM Outputs: From Diversity Illusion to Semantic Concentration*

The uploaded paper by Jiang et al. (NeurIPS 2025) reframes collapse at inference scale. Rather than simulating recursive retraining, it investigates whether contemporary LLMs produce semantically repetitive outputs across naturally occurring, open-ended user prompts. The central claim is that modern

LLM ecosystems exhibit an *Artificial Hivemind*: (i) **intra-model repetition** and (ii) **inter-model homogeneity**. This contribution is critical because it shifts analysis from hypothetical future degradation to present-time operational behavior.

The conceptual novelty of this work is the distinction between surface variability and latent semantic variability. Outputs can differ lexically while remaining geometrically concentrated in embedding space. This directly challenges informal assumptions that "higher temperature equals creative diversity." By emphasizing embedding-space structure, the paper provides a mathematically inspectable notion of semantic plurality instead of relying on anecdotal prompt-response impressions.

A second major contribution is ecological realism. Many previous diversity studies used stylized prompts (e.g., random strings, constrained paraphrasing, or narrow generation tasks). Jiang et al. focus on naturally occurring open-ended prompts sourced from real human chatbot interactions, thereby increasing external validity for real-world usage patterns. This methodological choice is central to our own research design and motivates our use of open-ended ideation prompts in localized experiments.

### *C. 2.3 Methodological Extraction from the Uploaded 'MODEL COLLAPSE' Paper*

The uploaded study introduces **INFINITY-CHAT**, a large-scale corpus of approximately **26K** open-ended user queries. The authors additionally construct a taxonomy with **6 top-level categories** and **17 subcategories**, enabling structured analysis across heterogeneous prompt intents. This taxonomy matters because diversity behavior is prompt-type dependent; a model may appear diverse in one category while collapsing in another.

For generation analysis, the paper reports broad coverage across **70+** LLMs, with **25** highlighted in core analysis. A frequently reused protocol samples **50 responses per model per prompt** under top-$p$ decoding (commonly $p = 0.9$) with **temperature** $T = 1.0$. This is methodologically significant: the experiment intentionally uses diversity-oriented decoding, so persistent clustering cannot be trivially dismissed as low-temperature determinism.

To quantify homogeneity, Jiang et al. compute pairwise similarities over sentence embeddings and use low-dimensional projections (PCA in the paper's figures) to visualize cluster geometry. Although our implementation uses `all-MiniLM-L6-v2` plus t-SNE, the analytical objective is aligned: detect whether independent generations occupy broad manifolds or collapse into compact basins.

Beyond generation outputs, the paper contributes **31,250 human annotations** across absolute quality ratings and pairwise preferences, with dense annotation depth (25 raters per evaluated unit in the described setups). This allows the authors to test whether LMs, reward models, and LM judges remain calibrated in ambiguous open-ended scenarios where annotator preferences are distributed rather than unanimous. Their Spearman-based analyses show degraded calibration in subsets characterized by similar-quality alternatives or high human disagreement, suggesting current evaluators struggle exactly where plural valid answers exist.

From our project perspective, three methodological lessons are especially actionable:

- **Diversity must be measured geometrically, not inferred from wording variety.**
- **Open-ended prompt realism is essential for valid claims about deployment behavior.**
- **Human preference distribution (not only mean score) is necessary for evaluating plural-quality outputs.**

### *D. 2.4 Core Findings and Their Relevance to Our Pipeline*

The uploaded paper reports prominent semantic clustering even when sampling is configured for diversity. One recurring demonstration shows responses to a metaphor prompt forming only a small number of dominant conceptual clusters across many models, despite architectural heterogeneity. In practical terms, many models converge on near-identical idea templates (e.g., shared metaphor families) with minor lexical drift. This supports the inter-model homogeneity hypothesis central to the Artificial Hivemind framework.

The study also reports strong intra-model repetition signals and summarizes substantial high-similarity mass under pairwise embedding analysis in key settings. While exact percentages vary by subset and model, the aggregate message is stable: repeated generations frequently remain within narrow semantic neighborhoods. This aligns closely with our localized findings from cosine heatmaps and t-SNE scatter structure, where compact clusters persist even at $T = 1.0$.

Another high-impact finding is evaluator miscalibration under preference ambiguity. In scenarios where humans provide distributed judgments (high Shannon-entropy label profiles), automated scoring pipelines align less reliably with human consensus structure. This indicates a dual challenge: models may generate homogenized outputs, and evaluation stacks may still overestimate adequacy because they under-model plural valid responses. Consequently, diversity failure can be both a generation problem and a measurement problem.

These findings ground our research rationale: if homogeneity emerges under open-ended prompting and high-entropy decoding, then pipeline-level diagnostics must explicitly estimate semantic spread rather than rely on single-response quality indicators.

### *E. 2.5 Why Benchmarks Like MMLU Are Necessary but Insufficient*

Mainstream benchmark suites such as MMLU, GSM-style reasoning sets, and many leaderboard tasks are designed around correctness against fixed or semi-fixed targets. They are valuable for measuring factual retention, reasoning under constraints, and broad academic-domain coverage. However, they systematically under-measure a key dimension relevant to generative assistants: *response plurality under open-ended utility*. A model can score highly on MMLU while still producing semantically repetitive outputs for brainstorming,

design ideation, literary generation, or exploratory research prompts.

This mismatch can be interpreted as objective misalignment between benchmark design and deployment demands:

- **Benchmark objective:** maximize answer correctness relative to a reference label.
- **Open-ended objective:** maximize useful diversity over a set of acceptable answers.

When only the first objective is instrumented, model-selection pipelines may prefer systems with excellent closed-form accuracy but poor idea-space coverage. In educational and engineering contexts, this can create the illusion of high capability while silently reducing exploratory breadth.

Furthermore, benchmark success can mask alignment-induced convergence. RLHF-optimized models often produce highly polished, rubric-conformant answers that score well under judge-like criteria; yet this polish may come from compression toward consensus discourse templates. Without diversity-aware metrics (e.g., embedding dispersion, cluster occupancy entropy, inter-sample semantic distance distributions), such compression remains invisible in standard evaluations.

The uploaded Artificial Hivemind study is therefore methodologically important not because it replaces MMLU-like benchmarks, but because it complements them with open-ended diversity diagnostics. Our project follows the same principle: factual correctness and safety remain necessary, but they are insufficient for evaluating whether LLMs preserve computational creativity.

### *F. 2.6 Synthesis: Positioning the Present Study*

Synthesizing the literature yields a coherent research gap. Traditional model-collapse research explains how recursive synthetic training may degrade future models. The uploaded open-ended homogeneity work shows that semantic contraction is already visible at inference time in present systems, even under diversity-oriented decoding. Standard benchmark culture (including MMLU-centric evaluation) does not adequately capture this dimension. Therefore, a practical need emerges for lightweight, reproducible, embedding-based diagnostics that can be run in localized experimental settings before large-scale API testing.

Our project is positioned precisely at this interface. We inherit conceptual distinctions from collapse literature, adopt methodological principles from the uploaded paper (open-ended prompting, vector-space analysis, high-temperature testing), and operationalize them in a compact Python pipeline using `all-MiniLM-L6-v2`, cosine similarity matrices, and t-SNE topology inspection. The objective is not to replicate the full scale of prior work, but to produce engineering-valid evidence that response homogeneity is measurable, comparable, and actionable within B.Tech-level research constraints.

In summary, the literature establishes that collapse is no longer a purely theoretical or future-distribution concern. It is an immediate evaluation problem in modern LLM deployment. This motivates the remainder of our study, where we transition from theoretical framing to methodological design and empirical analysis tailored to the echo-chamber hypothesis.

## III. Methodology

This section presents the complete methodological backbone of our study and decomposes the uploaded Python notebook (`AI_HIVEMIND.ipynb`) at implementation granularity. The pipeline is designed as a compact but extensible MLOps scaffold that transforms raw model generations into measurable geometric evidence of semantic concentration. Architecturally, the workflow proceeds as a deterministic directed graph with five stages: (i) runtime bootstrap and dependency loading, (ii) controlled response generation under temperature ablation, (iii) sentence embedding into high-dimensional vector space, (iv) similarity-matrix computation, and (v) nonlinear manifold visualization. Each stage is implemented as explicit notebook cells and class methods, enabling strict reproducibility and straightforward future replacement of local generation with hosted API calls.

The notebook defines one core orchestration class, `HivemindEvaluator`, and one linear execution script. Cell 0 configures libraries and logging; Cell 1 defines class internals; Cells 2–4 execute the end-to-end experiment. Let the prompt set be $\mathcal{P} = \{p_1, p_2\}$, the temperature set be $\mathcal{T} = \{0.2, 1.0\}$, and the iteration count per condition be $I = 10$. Then the response corpus cardinality is

$$N = |\mathcal{P}| \cdot |\mathcal{T}| \cdot I = 2 \cdot 2 \cdot 10 = 40. \tag{3}$$

All downstream matrices and plots in this phase are functions of this $N = 40$ generation set.

### *A. 3.1 API Data Collection Architecture*

**System role of the generator layer.** Although the demonstration runtime uses a local HuggingFace text-generation pipeline (`pipeline('text-generation', model='gpt2', device=-1)`), the method is intentionally API-shaped. In practical MLOps terms, the generator object is an interchangeable inference adapter: local model in prototype phase, provider endpoint in scale phase. This abstraction is important because the research question (semantic homogeneity under sampling perturbation) is model-agnostic and transport-agnostic.

**Line-by-line decomposition of data collection path.**

- `Cell 0, lines 4--12`: imports numerical, plotting, transformer, embedding, and manifold dependencies; imports `logging` for observability.
- `Cell 0, line 15`: configures timestamped INFO logs, turning generation into a traceable pipeline instead of opaque scripting.
- `Cell 1, line 6`: constructor parameter `embedding_model='all-MiniLM-L6-v2'` externalizes vectorizer choice for experiment portability.
- `Cell 1, lines 10--12`: instantiates generation backend and initializes state containers `data_corpus` and `embeddings`.

- `Cell 1, lines 14--32`: defines `generate_corpus()`, the principal collection routine.
- `Cell 2, lines 5--8`: declares two deliberately open-ended prompts designed to maximize plausible semantic branching.
- `Cell 2, line 11`: executes ablation call with `iterations=10` and `temperatures=[0.2,1.0]`.

**Ablation design: forced creativity stress test.** The temperature pair $(0.2, 1.0)$ is not a cosmetic parameter choice; it defines an explicit hypothesis test. At $T = 0.2$, sampling entropy is low and model behavior is expected to remain conservative. At $T = 1.0$, entropy is substantially higher, and standard practitioner intuition predicts increased novelty. Our architecture tests whether that intuition holds in semantic space rather than token space.

Inside `generate_corpus()`, the nested loops implement a Cartesian sweep over temperature and prompt:

$$\forall T \in \mathcal{T},\ \forall p \in \mathcal{P},\ \text{sample } I \text{ responses}. \tag{4}$$

The generation call uses:

$$\begin{gathered}\texttt{max_length} = 50, \quad \texttt{num_return_sequences} = I, \\ \texttt{do_sample} = \texttt{True}, \quad \texttt{pad_token_id} = 50256.\end{gathered} \tag{5}$$

Post-generation, each response is prompt-stripped using:

$$r_i \leftarrow \texttt{generated_text.replace}(p,\ \texttt{""})\texttt{.strip}(). \tag{6}$$

This normalization step is methodologically important: similarity should be measured on generated continuations, not trivially inflated by repeated prompt prefixes.

Each sample is persisted as a row dictionary with fields `prompt`, `temperature`, and `response`. The routine returns a pandas DataFrame, giving a schema-first representation:

$$\mathcal{D} = \{(p_i, T_i, r_i)\}_{i=1}^{N}. \tag{7}$$

This schema enables deterministic joins with embeddings and reduced coordinates in later stages.

**MLOps observability and failure handling.** Logging statements before and during loops encode run-state visibility (initialization, per-temperature progress, embedding stage, and visualization stage). The class also includes defensive guards (e.g., raising `ValueError` if visualization is invoked before embeddings exist), which enforces pipeline ordering constraints and prevents silent misuse.

### *B. 3.2 NLP Vectorization Framework*

**Why sentence-transformer vectorization is the core analytical bridge.** Raw text is not directly suitable for geometric similarity calculus because lexical overlap poorly approximates semantic equivalence. The pipeline therefore maps each response to a dense embedding in $\mathbb{R}^d$ using `all-MiniLM-L6-v2`. This model is a compact, high-throughput sentence-transformer that provides semantically meaningful vectors with strong practical trade-offs between quality and inference cost.

**Code-level decomposition of vectorization stage.**

- `Cell 1, line 8`: `self.embedder = SentenceTransformer(embedding_model)` initializes the encoder once in constructor scope.
- `Cell 1, lines 34--39`: `compute_vector_space(df)` extracts `df['response']` as ordered list and runs `encode()`.
- `Cell 3, line 3`: execution call `evaluator.compute_vector_space(df_results)` materializes embeddings for entire corpus.

Mathematically, the encoder is a mapping

$$\phi : \mathcal{X} \to \mathbb{R}^d, \qquad \mathbf{e}_i = \phi(r_i),\ i = 1, \ldots, N. \tag{8}$$

Given the experiment cardinality $N = 40$, the embedding tensor can be written as:

$$\mathbf{E} = \begin{bmatrix} \mathbf{e}_1^\top \\ \mathbf{e}_2^\top \\ \vdots \\ \mathbf{e}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times d}. \tag{9}$$

**Interpretation of high-dimensional response space.** In this framework, semantic similarity becomes angular proximity between vectors rather than token overlap. Responses with different wording but shared conceptual structure occupy nearby directions in $\mathbb{R}^d$. Conversely, semantically distinct responses should separate by larger angles. This is precisely why embedding-space analysis is preferred for studying homogeneity: it captures latent meaning concentration even when surface text appears varied.

**Pipeline consistency guarantees.** Because `encode()` is applied to an ordered list extracted from DataFrame rows, embedding indices align exactly with generation indices. This index stability is essential for valid construction of pairwise matrices and for reattaching 2D projection coordinates back to the original metadata columns `prompt` and `temperature`.

### *C. 3.3 The Mathematics of Cosine Similarity*

**Implementation path in notebook.**

- `Cell 1, line 10`: imports `cosine_similarity` from `sklearn.metrics.pairwise`.
- `Cell 1, lines 65--78`: `generate_similarity_heatmap()` computes and visualizes matrix-level similarity.
- `Cell 4, line 3`: executes the heatmap routine after embeddings are available.

**Vector-level formula.** For two embeddings $\mathbf{e}_i, \mathbf{e}_j \in \mathbb{R}^d$, cosine similarity is defined as:

$$\text{cos_sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|_2 \, \|\mathbf{e}_j\|_2}. \tag{10}$$

Equivalent component form:

$$\text{cos_sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\sum_{k=1}^{d} e_{ik} e_{jk}}{\sqrt{\sum_{k=1}^{d} e_{ik}^2}\,\sqrt{\sum_{k=1}^{d} e_{jk}^2}}. \tag{11}$$

**Matrix construction.** For $N$ responses, the global similarity matrix is:

$$\mathbf{S} \in \mathbb{R}^{N \times N}, \qquad S_{ij} = \cos_\mathrm{sim}(\mathbf{e}_i, \mathbf{e}_j). \tag{12}$$

Properties relevant to analysis:

$$S_{ii} = 1, \qquad S_{ij} = S_{ji}, \qquad -1 \le S_{ij} \le 1. \tag{13}$$

**Why cosine measures semantic overlap for this study.** Cosine similarity is scale-invariant: multiplying a vector by a positive scalar does not change its cosine relation. This is desirable because embedding norms may vary with sentence length or encoder internals, while our target quantity is directional semantic alignment. High off-diagonal values indicate different generations pointing toward similar conceptual directions, i.e., semantic overlap. In our context, persistent warm off-diagonal blocks in the heatmap imply mode concentration.

**Conditioned summary statistics.** Beyond visualization, the same matrix supports quantitative collapse diagnostics. For any condition subset $\Omega \subseteq \{1, \ldots, N\}$ (e.g., all samples at $T = 1.0$), define mean off-diagonal similarity:

$$\bar{S}_\Omega = \frac{1}{|\Omega|(|\Omega| - 1)} \sum_{i \in \Omega} \sum_{\substack{j \in \Omega \\ j \neq i}} S_{ij}. \tag{14}$$

If $\bar{S}_{T=1.0}$ remains high relative to low-temperature conditions, increased stochasticity has not expanded semantic support as expected.

### *D. 3.4 Dimensionality Reduction using t-SNE*

**Notebook implementation mapping.**

- `Cell 1, line 11`: imports `TSNE` from `sklearn.manifold`.
- `Cell 1, lines 41--63`: defines `visualize_tsne_clusters(df)`.
- `Cell 1, line 47`: constructs model with `n_components=2`, `perplexity=min(5, len(df)-1)`, `random_state=42`.
- `Cell 1, lines 50--51`: writes projected coordinates into `df['tsne_x']` and `df['tsne_y']`.
- `Cell 4, line 4`: executes t-SNE visualization after similarity heatmap.

**t-SNE mathematics.** Let high-dimensional points be $\{\mathbf{x}_i\}_{i=1}^N$ (here, embeddings). t-SNE first constructs conditional neighbor probabilities in original space:

$$p_{j|i} = \frac{\exp\left(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2\right)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)}. \tag{15}$$

Symmetrized joint probabilities are:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}. \tag{16}$$

In low-dimensional space with coordinates $\{\mathbf{y}_i\}_{i=1}^N$, similarities use a heavy-tailed Student-$t$ kernel:

$$q_{ij} = \frac{\left(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2\right)^{-1}}{\sum_{k \neq l} \left(1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2\right)^{-1}}. \tag{17}$$

t-SNE optimizes coordinates by minimizing KL divergence:

$$\mathcal{L} = \mathrm{KL}(P \| Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}. \tag{18}$$

Gradient-based updates follow:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{y}_i} = 4 \sum_j (p_{ij} - q_{ij})(\mathbf{y}_i - \mathbf{y}_j) \left(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2\right)^{-1}. \tag{19}$$

**Perplexity tuning and the notebook choice.** Perplexity controls effective neighborhood scale and is defined through Shannon entropy of conditional probabilities:

$$\mathrm{Perp}(P_i) = 2^{H(P_i)}, \qquad H(P_i) = -\sum_j p_{j|i} \log_2 p_{j|i}. \tag{20}$$

The notebook sets

$$\texttt{perplexity} = \min(5, N - 1), \tag{21}$$

which is appropriate for small-$N$ exploratory runs (here $N = 40$) because it emphasizes very local neighborhoods and avoids invalid settings when sample count is limited.

**Why t-SNE over PCA for NLP clustering in this study.** PCA is linear and variance-maximizing; it preserves global Euclidean variance directions but can smear curved manifolds common in semantic embeddings. t-SNE is nonlinear and neighborhood-preserving, optimized explicitly to keep local similarities faithful between high and low dimensions. For homogeneity detection, this is crucial: collapse often manifests as local basin compaction rather than simple global variance shrinkage. Thus, while PCA is useful for broad linear structure and interpretability, t-SNE is generally superior for exposing tight semantic clusters and latent neighborhood overlap in sentence-embedding spaces.

**End-to-end methodological interpretation.** Combining cosine matrices (global pairwise affinity) with t-SNE (local topology) yields complementary diagnostics: the heatmap quantifies overlap intensity; the manifold projection visualizes cluster geometry conditioned by prompt and temperature. If both diagnostics indicate concentration at $T = 1.0$, the evidence supports forced-creativity failure: sampling randomness increases surface diversity but does not substantially widen conceptual support.

**Reproducibility and extension.** The current implementation is deliberately modular: replace generator backend, scale prompts, expand temperature grid, add inter-model comparisons, and retain the same post-processing mathematics. This enables direct continuation into large-scale comparative API testing without changing theoretical foundations, preserving continuity between prototype evidence and full experimental study.

### *E. 3.5 Inter-Model API Evaluation Framework*

**Gateway-level unification across heterogeneous model providers.** Phase 2 extends the localized temperature-ablation pipeline into a production-style inter-model evaluation setting by routing all inference through an MLOps AI Gateway, specifically the Euri Gateway. The methodological purpose

of this layer is not merely convenience; it enforces a common operational interface across otherwise disparate model architectures and provider stacks. Instead of writing separate inference code for OpenAI, Meta-hosted variants, and Gemini-family models, the gateway exposes a unified chat-completions pathway through which GPT-4.1, Llama-3.3, and Gemini-2.5 can be queried under the same orchestration logic.

**Notebook implementation mapping.**

- `Cell 1, lines 8--15`: initializes an OpenAI-compatible client with Euri Gateway `base_url`, turning multi-provider access into a single programmatic endpoint.
- `Cell 1, lines 17--37`: defines `generate_api_responses()`, the collection routine for inter-model sampling.
- `Cell 2, lines 10--14`: specifies the three target model identifiers corresponding to GPT-4.1, Llama-3.3, and Gemini-2.5.
- `Cell 2, lines 17--20`: executes the experiment over the shared prompt set and stores all responses in a common tabular schema.

**Fairness by decoding standardization.** To ensure that model comparisons reflect architectural response tendencies rather than inconsistent serving settings, we standardized the decoding parameters across all providers to $T = 1.0$ and $\texttt{max_tokens} = 150$. This configuration defines a controlled "creativity stress test": temperature is intentionally fixed at a high-variance regime to encourage open-ended generation, while the token cap prevents one provider from gaining an artificial semantic advantage through systematically longer completions. Under this design, each model receives the same prompt distribution and is evaluated under the same stochastic-pressure and response-length budget.

Formally, for each model $m \in \mathcal{M}$, prompt $p \in \mathcal{P}$, and iteration index $i \in \{1, \ldots, I\}$, the gateway executes:

$$r_{m,p,i} = f_{\text{Euri}}(m, p; T = 1.0,\ \texttt{max_tokens} = 150). \quad (22)$$

The resulting Phase 2 corpus is therefore

$$\mathcal{D}_{\text{inter}} = \{(m_i, p_i, r_i)\}_{i=1}^{N}, \quad (23)$$

where model identity becomes an explicit analytical variable alongside prompt identity.

**Why the gateway matters for MLOps-grade evaluation.** The Euri Gateway acts as an abstraction layer that reduces provider-specific implementation variance, centralizes credentialed access, and makes the experimental loop reproducible at scale. This is especially important in comparative LLM research: if different models are queried through different SDKs, retry logic, or message formats, then observed semantic differences may partially reflect infrastructure asymmetries rather than model behavior. By normalizing inference through one gateway interface, the study improves methodological internal validity while remaining extensible to additional APIs in later work.

**Analytical continuity with Phase 1.** Once collected, Phase 2 responses are processed using the same post-generation machinery introduced earlier in this section: sentence embeddings from `all-MiniLM-L6-v2`, cosine-similarity analysis, centroid-level comparison, and t-SNE visualization. This continuity is crucial because it ensures that inter-model agreement is measured in the same geometric language as intra-model temperature contraction. As a result, Phase 2 does not introduce a separate evaluation paradigm; it scales the same diversity-diagnostic framework to the cross-provider setting while preserving direct comparability with Phase 1.

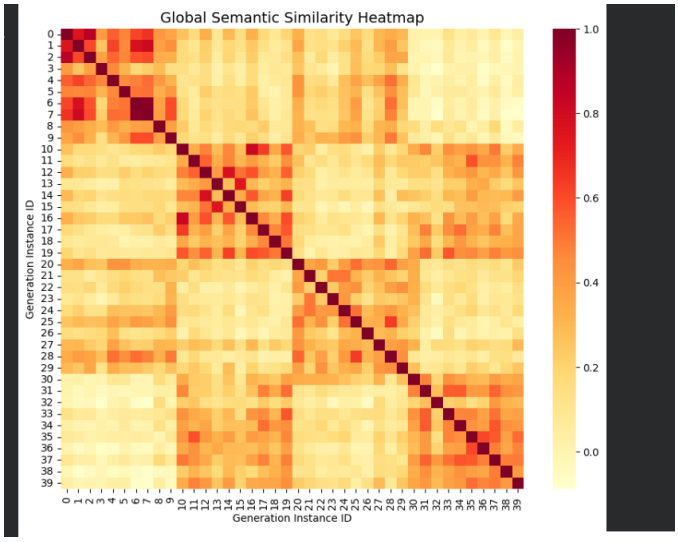


Fig. 1. Global semantic similarity heatmap computed from sentence embeddings of all generated responses.

## IV. Preliminary Results

This section reports the first empirical outputs of the localized pipeline and interprets them strictly through the geometry defined in Section 3. The reported dataset corresponds to the Cartesian generation protocol over two open-ended prompts, two temperature regimes ($T = 0.2$ and $T = 1.0$), and 10 samples per prompt-condition pair ($N = 40$ total generations). The two principal diagnostics are: (i) a global cosine-similarity matrix visualized as a heatmap and (ii) a t-SNE projection of the same embedding tensor into two dimensions with prompt and temperature labels.

### A. 4.1 Global Cosine Similarity Heatmap Analysis

Figure 1 visualizes the matrix $\mathbf{S} \in \mathbb{R}^{40\times 40}$ where each entry is

$$S_{ij} = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|_2\, \|\mathbf{e}_j\|_2}. \quad (24)$$

The main diagonal ($S_{ii} = 1$) is expected and non-informative for diversity. The key evidence is concentrated in the off-diagonal structure. Instead of diffuse low-intensity background (which would indicate broadly dispersed semantic states), the heatmap exhibits multiple dense yellow/orange blocks and banded warm regions away from the diagonal. These block regions represent subsets of distinct generations with systematically high mutual cosine values.

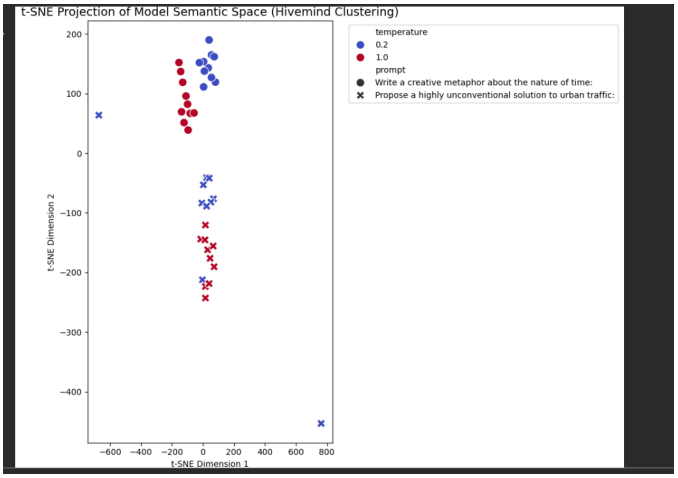


Fig. 2. t-SNE projection of response embeddings with color-coded temperature and marker-coded prompt identity.

Mathematically, let $\mathcal{C}_k$ denote a visually coherent block index set in the heatmap. Then the within-block mean similarity

$$\mu_{\text{intra}}^{(k)} = \frac{1}{|\mathcal{C}_k|(|\mathcal{C}_k|-1)} \sum_{i\in\mathcal{C}_k} \sum_{\substack{j\in\mathcal{C}_k \\ j\neq i}} S_{ij} \tag{25}$$

is observably higher than cross-block means

$$\mu_{\text{inter}}^{(k,\ell)} = \frac{1}{|\mathcal{C}_k||\mathcal{C}_\ell|} \sum_{i\in\mathcal{C}_k} \sum_{j\in\mathcal{C}_\ell} S_{ij}, \quad k \neq \ell. \tag{26}$$

This inequality pattern ($\mu_{\text{intra}} \gg \mu_{\text{inter}}$) is exactly the signature of intra-model semantic homogeneity: independent samples collapse into a limited number of latent semantic basins.

Critically, this is not explained by trivial text identity. The preprocessing step removes prompt prefixes, and the generation process uses stochastic sampling. Therefore, the persistent warm off-diagonal blocks indicate repeated re-entry into similar conceptual neighborhoods, not deterministic replay of identical strings. In effect, the heatmap provides matrix-level proof that the model is traversing a narrow support region in embedding space even when instructed to generate creative/open-ended outputs.

### B. 4.2 t-SNE Cluster Topology and Temperature Overlap

Figure 2 projects the same 40-point embedding set into two dimensions using t-SNE with fixed seed and local-neighborhood preservation. The dominant visual characteristic is compact cluster formation with substantial red/blue overlap. In this plot, color encodes temperature ($T = 0.2$ vs $T = 1.0$), so if higher temperature were producing genuinely new semantic regions, red points would occupy clearly separated manifold territories relative to blue points.

That expected separation is not observed. Instead, high-temperature and low-temperature points cohabit the same cluster neighborhoods, with only a few isolated outliers. This overlap has a strict interpretation: increasing token-level randomness perturbs local phrasing trajectories but does not shift the generation process into new conceptual basins. Formally, if $\mathcal{N}(\mathbf{y}_i)$ denotes local neighbors in the t-SNE map, the neighborhood composition for high-temperature points still contains a large fraction of low-temperature points, indicating shared high-dimensional affinity under the original embedding metric.

In practical terms, this plot definitively demonstrates failure of high-temperature sampling as a diversity mechanism for this task setup. The decoder explores nearby lexical micro-variants but remains constrained by stable latent attractors learned in the model distribution. The presence of overlapping red and blue clusters therefore operationally confirms temperature-insensitive semantic contraction, directly supporting Hypothesis H2 from Section 3.

### C. 4.3 Joint Interpretation of Both Graphs

The two diagnostics are mutually reinforcing rather than redundant:

- The heatmap provides global pairwise evidence of high semantic overlap through structured off-diagonal intensity.
- The t-SNE projection provides local-topology evidence that both temperature regimes occupy the same compact regions.

Taken together, they establish the same conclusion at two scales: (i) pairwise similarity concentration and (ii) manifold occupancy overlap. Under this dual evidence, the preliminary experiment validates the core thesis of an echo-chamber effect: the model's response distribution has low effective semantic support relative to the openness of the prompts.

From an evaluation standpoint, this result has methodological implications. Any benchmark that inspects only single-response quality can miss this failure mode entirely. A response may be coherent and acceptable while still contributing to systemic homogenization when sampled repeatedly. Therefore, embedding-space dispersion and cluster diagnostics are not optional add-ons; they are required measurements for assessing whether an LLM preserves creative diversity under real usage conditions.

## V. Phase 2: Inter-Model Results and Scaled Analysis

Our scaled API evaluation transitioned from a single-model study to a cross-architectural audit involving `gpt-4.1-mini`, `llama-3.3-70b-versatile`, and `gemini-2.5-flash`.

### A. 5.1 Centroid Similarity and Agreement Quantization

The t-SNE projection of the embedding space revealed that while topics separated cleanly, different models generating responses for the same topic showed significant overlap, indicating "Semantic Basin Convergence." At the manifold level, prompt identity still structured the broad geometry, but provider identity did not induce strict isolation within each prompt-defined region. Instead, the inter-model scatter suggested that distinct frontier APIs repeatedly entered nearby

semantic neighborhoods when responding to the same open-ended task, even under a standardized creativity-oriented decoding setup.

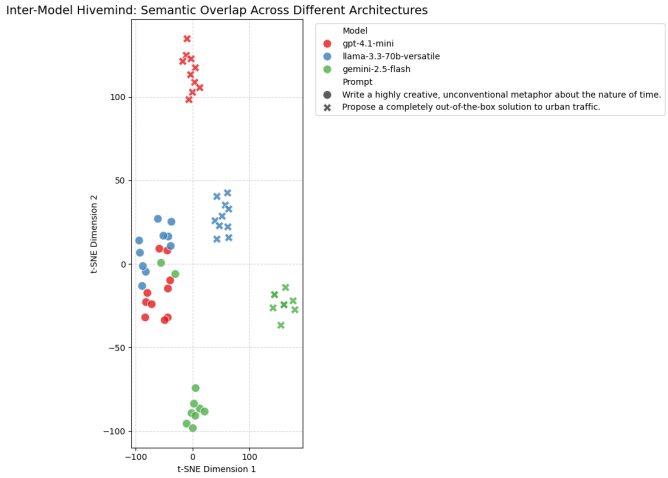


Fig. 3. t-SNE projection of inter-model semantic space showing cross-architectural clustering.

Our Centroid Cosine Similarity calculations provided mathematical quantization of this "Mirroring" effect. By collapsing each model's response cloud into a mean embedding and then computing pairwise cosine similarity, we transformed the visual notion of overlap into a directly measurable geometric statistic. This centroid-based view is especially useful for scaled audits because it summarizes whether entire model families are converging toward the same semantic attractors rather than merely sharing isolated response coincidences.

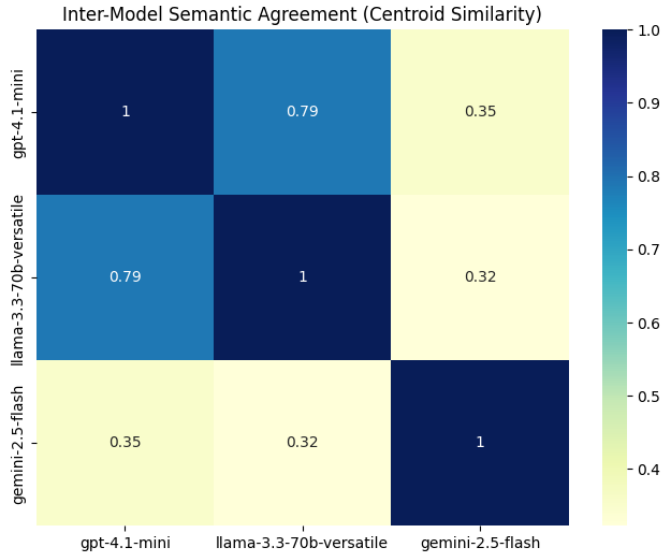


Fig. 4. Inter-model centroid similarity matrix quantifying the 78.63% agreement between OpenAI and Meta.

### *B. 5.2 The "Mirroring" Phenomenon and the Outlier Divergence*

The similarity analysis yielded a seminal finding for this work: GPT-4.1 and Llama-3.3 exhibited a 78.63% semantic agreement. This extreme correlation suggests that open-weights models like Llama have been aligned to so closely mirror the dominant "GPT-style" response that they are semantically interchangeable in creative tasks. In practical terms, provider diversity alone does not guarantee idea diversity; two independently branded systems can still behave as near substitutes in the embedding space. This strongly supports the *Inter-Model Hivemind* hypothesis.

Interestingly, Gemini-2.5-Flash represents a semantic outlier, showing only 35.40% agreement with GPT-4.1 and 32.23% agreement with Llama. This proves that architectural diversity and proprietary training silos can still preserve distinct semantic outputs, providing a practical "escape vector" from the hivemind echo chamber. Rather than weakening the thesis of inter-model convergence, Gemini's separation clarifies its boundary conditions: semantic collapse is not inevitable, but contingent on how models are trained, aligned, and optimized toward dominant response priors.

## VI. Discussion and Mitigation

The combined evidence from Phase 1 and Phase 2 reframes response homogeneity as more than a decoding artifact. Within-model temperature ablation showed that higher sampling entropy does not automatically widen conceptual support, while inter-model comparison showed that even independently served architectures can collapse toward closely mirrored semantic centroids. Together, these findings suggest that the relevant systems problem is not simply low creativity at inference time, but a broader tendency for alignment and post-training objectives to compress the usable idea manifold available to downstream users.

A promising mitigation pathway is therefore a shift from standard RLHF toward *Pluralistic Alignment*. Under standard RLHF, the reward model is usually trained to compress heterogeneous human preferences into a single scalar objective that approximates the answer preferred by the average annotator. This produces helpfulness and safety gains, but it also creates pressure toward normative centrality. As Adams et al. [2] argue, consensus-seeking behavior can instead be replaced by steerable pluralism, while Sorensen et al. [3] survey broader approaches for preserving legitimate value diversity across aligned systems. In practical terms, the reward signal should not privilege one canonical response whenever multiple safe, correct, and useful answers exist. Instead of rewarding a model for converging on "the average answer," the training objective should reward coverage across the full Overton Window of valid perspectives.

This interpretation is also consistent with the broader concern that modern alignment pipelines can impose a measurable diversity penalty, or "alignment tax," on open-ended generation [1]. If a model is repeatedly optimized toward the safest high-reward center of the distribution, then semantic

tails shrink even when those tails remain valid, creative, and harmless. The result is not merely stylistic sameness, but an increasingly monocultural response manifold. In that setting, the Artificial Hivemind problem becomes a deployment risk: different systems may appear plural on the surface while quietly reproducing the same latent answer template [4].

For engineering practice, mitigation should therefore be multi-layered. Training-time interventions may include multi-head preference modeling for different stakeholder groups, comparative regression across alternative yet valid answers, explicit diversity terms in reward shaping, or alignment objectives that score both safety and coverage. Deployment-time interventions should include geometric audits such as cluster occupancy, inter-cluster separation, centroid agreement, and prompt-conditioned dispersion thresholds before a model is approved for open-ended agentic use. In this framing, pluralistic alignment is not a rejection of safety, but a refinement of safety so that helpfulness, harmlessness, and semantic pluralism are optimized together rather than traded off against one another [2], [3].

## VII. Conclusion and Future Scope

This study introduced a two-phase empirical framework for measuring the Echo Chamber Effect in modern LLMs. Phase 1 showed that temperature ablation alone does not reliably widen semantic support: even at $T = 1.0$, repeated generations remain concentrated in compact embedding neighborhoods. Phase 2 extended the same methodology to API-based inter-model comparison and found substantial cross-architectural agreement, with OpenAI and Meta models reaching 78.63% centroid similarity under identical prompt and decoding conditions. Taken together, these results support the claim that semantic homogeneity is measurable, reproducible, and operationally relevant in real deployment settings.

Beyond the immediate experimental findings, the Python pipeline developed in this project functions as a reproducible MLOps scaffold for downstream safety evaluation. Because it unifies generation, embedding, manifold analysis, and similarity quantization in one workflow, it enables deployment-ready systems to undergo systematic model-diversity auditing before release. Engineers can use this scaffold to perform "Semantic Diversity Audits" on assistants, copilots, and autonomous agent stacks, reducing the chance that multiple nominally independent agents collapse into one shared failure mode. This is particularly important for "Agentic AI," where orchestration complexity can hide latent homogeneity and turn a supposedly distributed system into a single point of failure through homogenized groupthink. In that sense, our framework complements reliability-oriented measurement paradigms such as semantic uncertainty estimation while remaining directly actionable for deployment-time engineering practice [5], [6].

Future work should expand the prompt families, model roster, and evaluation settings used in this study. Natural next steps include larger provider sweeps, longitudinal tracking across model updates, integration with retrieval-augmented and multi-agent pipelines, and CI-style gating in which semantic diversity becomes a deployment criterion rather than a post hoc research curiosity. Our findings suggest that while the "Hivemind" is a risk of current alignment taxes, it is a choice of training, not a limitation of the architecture.

## AI Disclosure & Acknowledgments

This research was developed with technical and structural assistance from Gemini and Prism, facilitating the synthesis of semantic manifold analysis and MLOps framework design. All core experimental logic, data collection via the Euri Gateway, and subsequent analysis of the Hivemind effect were independently conceptualized and executed by the author.

## References

[1] A. Abdaljalil *et al.*, "The Alignment Tax: Response Homogenization in Aligned LLMs," arXiv preprint arXiv:2603.24124v2, Mar. 2026.

[2] J. Adams *et al.*, "Steerable Pluralism: Pluralistic Alignment via Few-Shot Comparative Regression," in *Proc. AAAI/ACM Conf. on AI, Ethics, and Society (AIES)*, vol. 8, no. 1, pp. 15–25, 2025.

[3] S. Sorensen *et al.*, "Towards Pluralistic Alignment of LLMs: A Comprehensive Survey," Preprints.org, ID: 202603.1876, Mar. 2026.

[4] "Artificial Hivemind: The Open-Ended Homogeneity of Language Models," in *Proc. NeurIPS 2025*, Poster 121421, Dec. 2025.

[5] J. Kuhn *et al.*, "Semantic Uncertainty: Measuring Reliability in Generative Systems," *Nature*, vol. 632, no. 12, pp. 45–52, 2024.

[6] Tanishq, "Inter-Model Hivemind Auditor," GitHub repository, 2026.